Aim: Using DecisionTreeClassifier and RandomForestClassifier.

1.Importing the important libraries.

```
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

2.Loading the dataset.

```
from pyodide.http import open_url
url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"
with open_url(url) as f:
    X = pd.read_csv(f)
```

3.Training and testing the dataset.

```
from sklearn.model_selection import train_test_split

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X[['Age', 'Fare']])

y = X['Survived']

x = X.drop(columns = ['Survived'])

x_train, x_test, y_train, y_test = train_test_split(X_scaled, y, test_size = 0.5, random_state = 42)
```

4.Finding accuracy and report using DecisionTreeClassifier.

```
dt = DecisionTreeClassifier(random_state = 42)
dt.fit(x_train, y_train)
y_pred_dt = dt.predict(x_test)
print("Decision Tree Accuracy:", accuracy_score(y_test, y_pred_dt))
print("Classification Report:\n", classification_report(y_test, y_pred_dt))
```

OUTPUT:

```
Decision Tree Accuracy: 0.6210762331838565
Classification Report:
               precision    recall  f1-score   support

           0       0.67      0.72      0.69       267
           1       0.53      0.47      0.50       179

    accuracy                           0.62       446
   macro avg       0.60      0.60      0.60       446
weighted avg       0.62      0.62      0.62       446
```

5.Finding accuracy and report using RandomForestClassifier.

```
rf = RandomForestClassifier(random_state = 42, n_estimators = 100)
rf.fit(x_train, y_train)
y_pred_rf = rf.predict(x_test)
print("Random Forest Accuracy:", accuracy_score(y_test, y_pred_rf))
print("Classification Report:\n", classification_report(y_test, y_pred_rf))
```

OUTPUT:

```
Random Forest Accuracy: 0.625560538116592
Classification Report:
               precision    recall  f1-score   support

           0       0.67      0.73      0.70       267
           1       0.54      0.46      0.50       179

    accuracy                           0.63       446
   macro avg       0.61      0.60      0.60       446
weighted avg       0.62      0.63      0.62       446
```